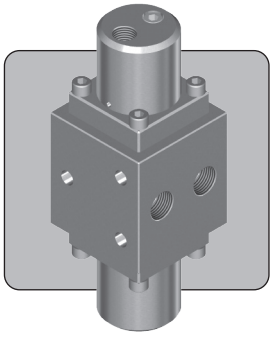
S-6B37

DIRECTIONAL CONTROL VALVE

3/8″ UP TO 690 BAR 10,000 PSI

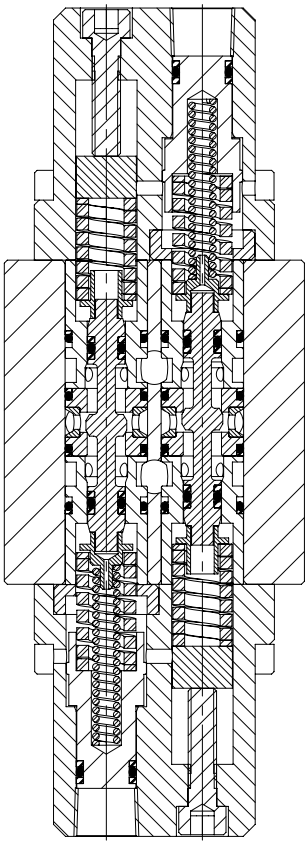


The S-6B37 is a 3/8″ ported 2 position 4 way directional control valve fitted with two hydraulic or pneumatic pilot actuators that are designed to be pressurised simultaneously.

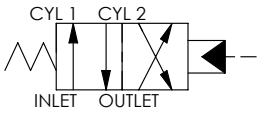
The valve uses a balanced spool and seat design such that pressure in the main lines has a negligible effect on the actuator force required to operate the valve.

The valve allows bi-directional flow and offers low leakage from a well proven design.

- 2 position 4 way directional control valve
- Often used to control double acting cylinders or actuators
- Balanced spool design, with spring return
- Wide pilot pressure range
- Vent chambers prevent mixing of main line and pilot fluid
- Subsea variants available, contact us for advice
- Suitable for use with mineral oils and water glycols
- Various seal options available
- Various porting options available



Specifications

BASIC MODEL NUMBER	S-6B37
SYMBOL	
MAX WORKING PRESSURE	690 bar (10,000 psi)
CV (FLOW CAPACITY)	0.44
FLUID	Liquids only See materials section
TEMPERATURE RANGE	See Product Selector opposite and Technical Data section
PORT SIZE	3/8″
WEIGHT	H - 3.25 kg (7.2 lb) A3 - 3.10 kg (6.8 lb)

Specifications may change without notice

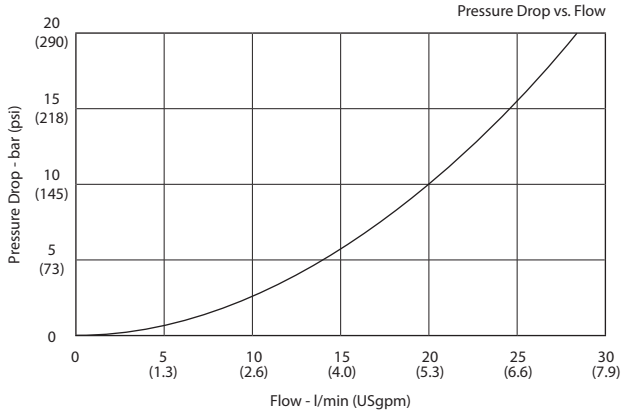
Materials

Externally Exposed Parts: 316 and other 300 and 400 series stainless steels with electroless nickel plated springs and bolts.

Internally Wetted Parts: 316, 431 and 440C stainless steel, acetal and PTFE. The standard valve is designed for use with non-corrosive liquids only, such as mineral oil and water glycol. Subsea variants are available containing materials offering improved corrosion resistance, contact us for advice.

⚠ The standard valve has Viton® seals. Further seal options are available via the Product Selector. Compatibility with the working fluid at the operating temperature must be considered.

Typical Performance

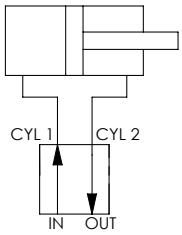
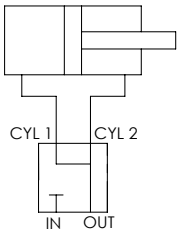
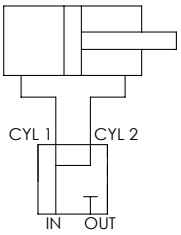
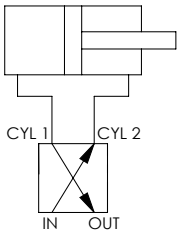


Typical performance based on water glycol

Actuator Options



The S-6B37 is available with a high pressure hydraulic or low pressure pneumatic pilot actuator option. The valve has two pilot ports and is primarily designed to be switched by pressurising both pilot ports simultaneously. The valve will begin to switch at approximately 30% of the minimum quoted pilot pressure. At the minimum quoted pilot pressure the valve will be fully switched and will achieve minimum leakage. Pressurising the pilot ports independently generates other flow configurations that can be useful in certain applications. The following table illustrates the available circuit configurations and how they would function when connected to a double acting cylinder or actuator, assuming that the inlet port is connected to a pressure source and the outlet port is connected to return.

Pilot Port 1	No Pressure	Pressurised	No Pressure	Pressurised
Pilot Port 2	No Pressure	No Pressure	Pressurised	Pressurised
Schematic				
Function	**Extend Mode** This is a standard straight flow path configuration, and causes the cylinder to extend	**Float Mode** This is a float configuration, where both sides of the cylinder are connected to return, allowing the cylinder to move when subject to external forces	**Regen Mode** This is a regenerative configuration, where both sides of the cylinder are connected to the pressure source. This causes the cylinder to extend faster because the flow out from the cylinder is added to the flow in from the pressure source. Note that the force available from the cylinder in this mode is reduced compared to standard extend mode, and that this functionality cannot be applied to double rod cylinders	**Retract Mode** This is a standard crossed flow path configuration, and causes the cylinder to retract

Product Selector

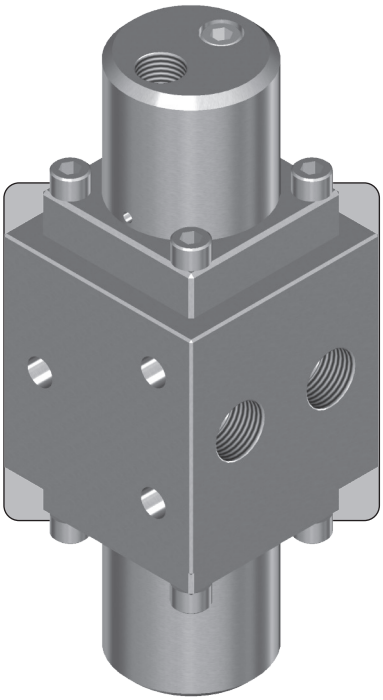
S-6B37 P - H - NBR - 10K

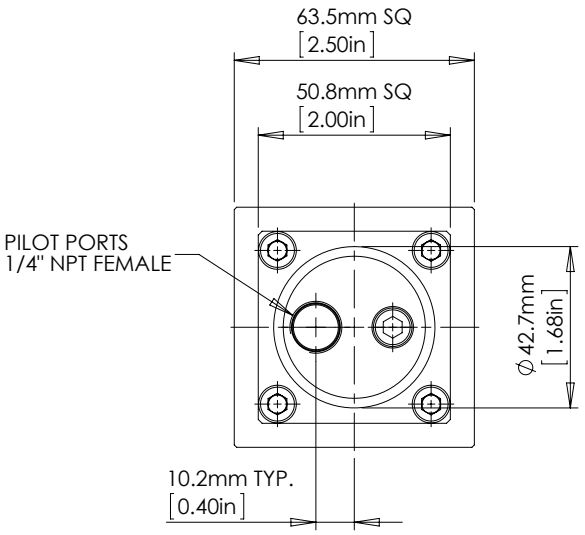
VALVE TYPE	PORTING OPTIONS	ACTUATOR OPTIONS	SEAL OPTIONS	MAX WORKING PRESSURE
S-6B37	P 3/8" BSPP female N 3/8" NPT female See the relevant installation drawing for pilot port type and size	H High pressure pilot 69 - 621 bar (1,000 - 9,000 psi) A3 Low pressure pilot 4.1 - 10.3 bar (60 - 150 psi)	Leave blank for Viton® seals -10˚C to +120˚C NBR Nitrile seals -20˚C to +120˚C KLZ Kalrez® seals -20˚C to +120˚C HNBR Hydrogenated Nitrile seals -20˚C to +120˚C Further seal options are available on request	10K 690 bar (10,000 psi) Liquid
Repair and seal kits are available on request For a seal kit add SK at the end of the model code and for a repair kit add RK		The A3 actuator is fitted with a Nitrile diaphragm regardless of seal option choice and actuator fluid temperature should not be allowed to exceed 100˚C		

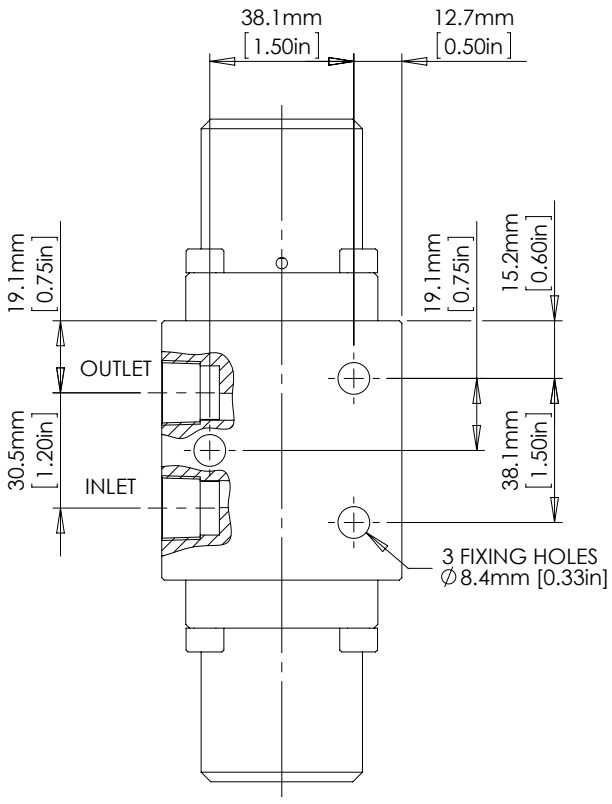
S-6B37

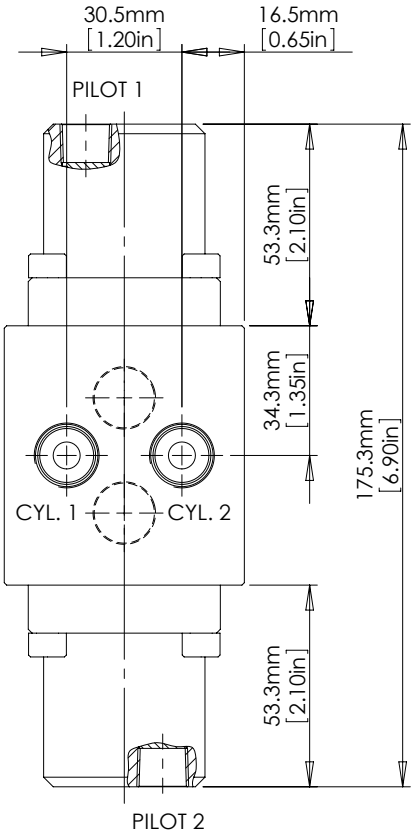
DIRECTIONAL CONTROL VALVE

S-6B37-H

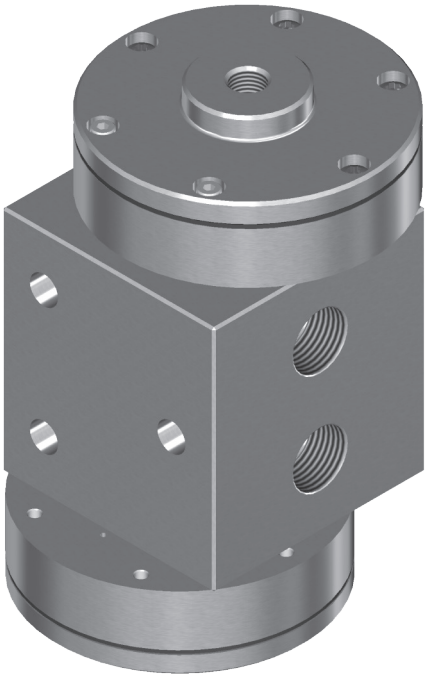


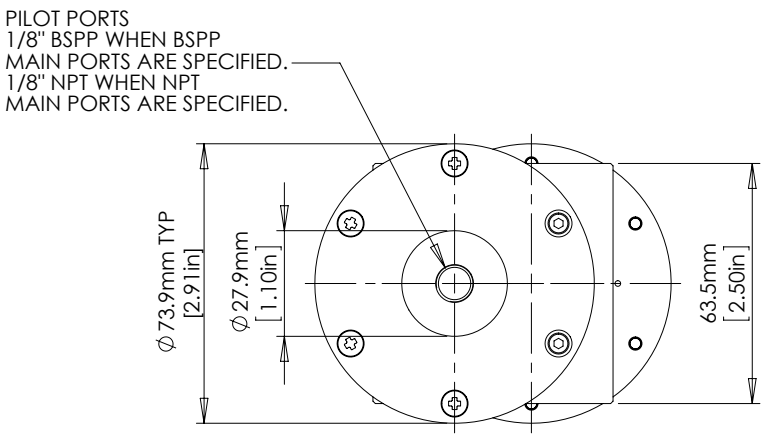


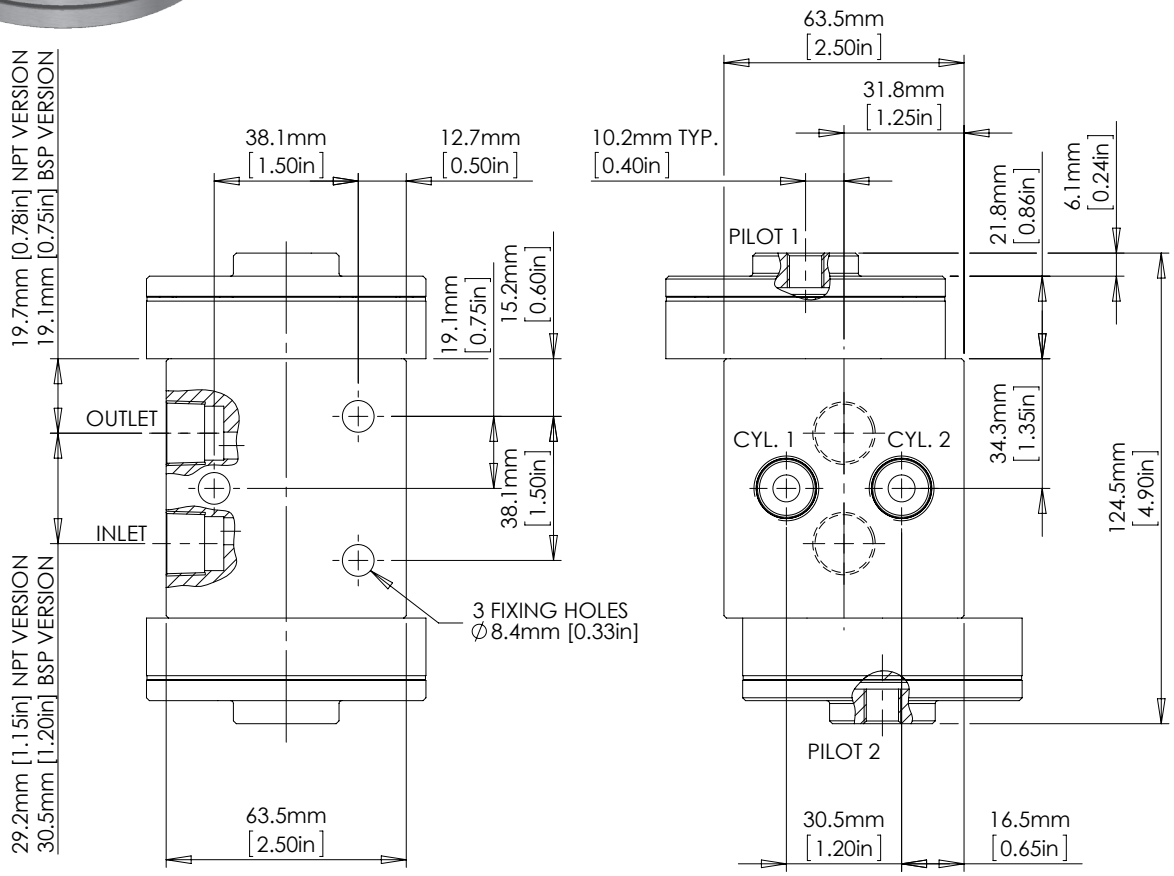




BS VALVES




PILOT PORTS
1/8" BSPP WHEN BSPP
MAIN PORTS ARE SPECIFIED.
1/8" NPT WHEN NPT
MAIN PORTS ARE SPECIFIED.
Ø73.9mm TYP
[2.91in]
Ø27.9mm
[1.10in]
63.5mm
[2.50in]


19.7mm [0.78in] NPT VERSION
19.1mm [0.75in] BSP VERSION
38.1mm
[1.50in]
12.7mm
[0.50in]
19.1mm
[0.75in]
15.2mm
[0.60in]
OUTLET
INLET
38.1mm
[1.50in]
3 FIXING HOLES
Ø8.4mm [0.33in]
29.2mm [1.15in] NPT VERSION
30.5mm [1.20in] BSP VERSION
63.5mm
[2.50in]
63.5mm
[2.50in]
31.8mm
[1.25in]
10.2mm TYP.
[0.40in]
PILOT 1
21.8mm
[0.86in]
6.1mm
[0.24in]
CYL. 1
CYL. 2
34.3mm
[1.35in]
124.5mm
[4.90in]
PILOT 2
30.5mm
[1.20in]
16.5mm
[0.65in]